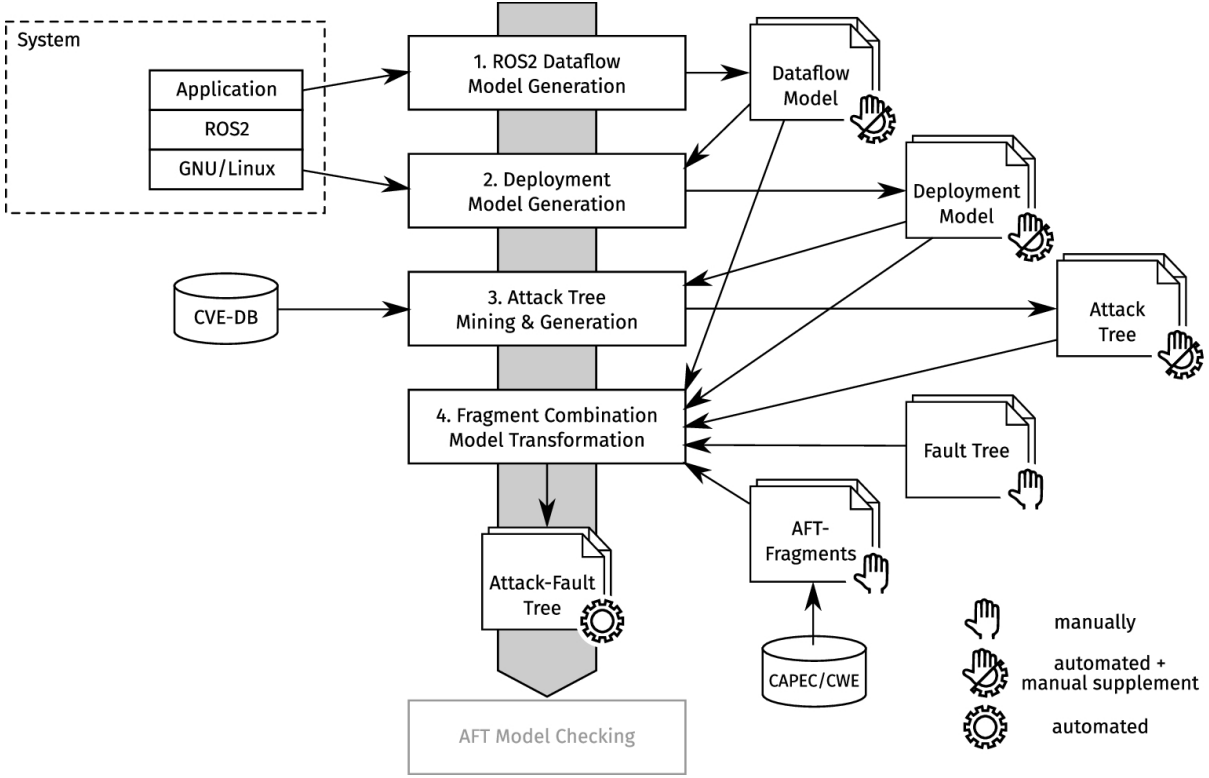
Combined Safety and Security Analysis in Self-Adaptive Systems

SAFESEC Workshop – Input Models and Model Generation

Overview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GNU/Linux
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2. Deployment Model Generation
3. Attack Tree Mining & Generation
4. Fragment Combination Model Transformation
CVE-DB
Dataflow Model
Deployment Model
Attack Tree
Fault Tree
AFT-Fragments
CAPEC/CWE
Attack-Fault Tree
AFT Model Checking
manually
automated + manual supplement
automated

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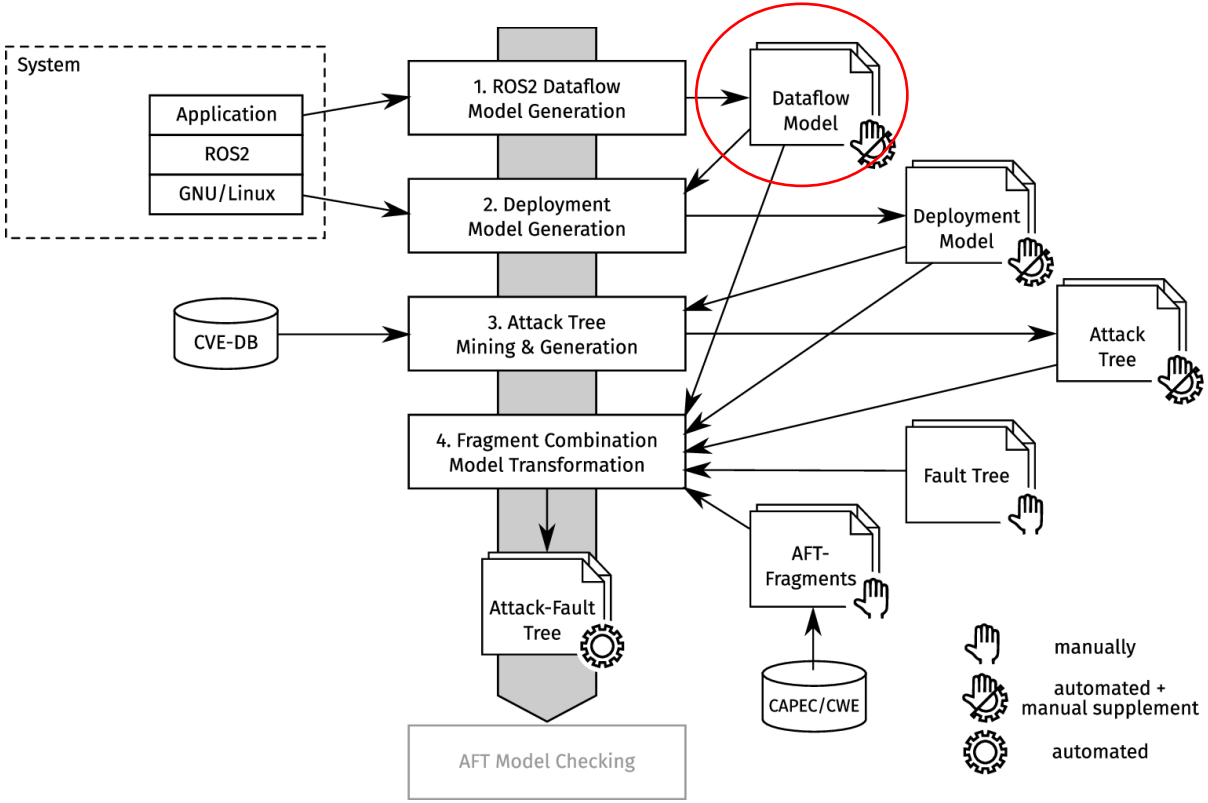
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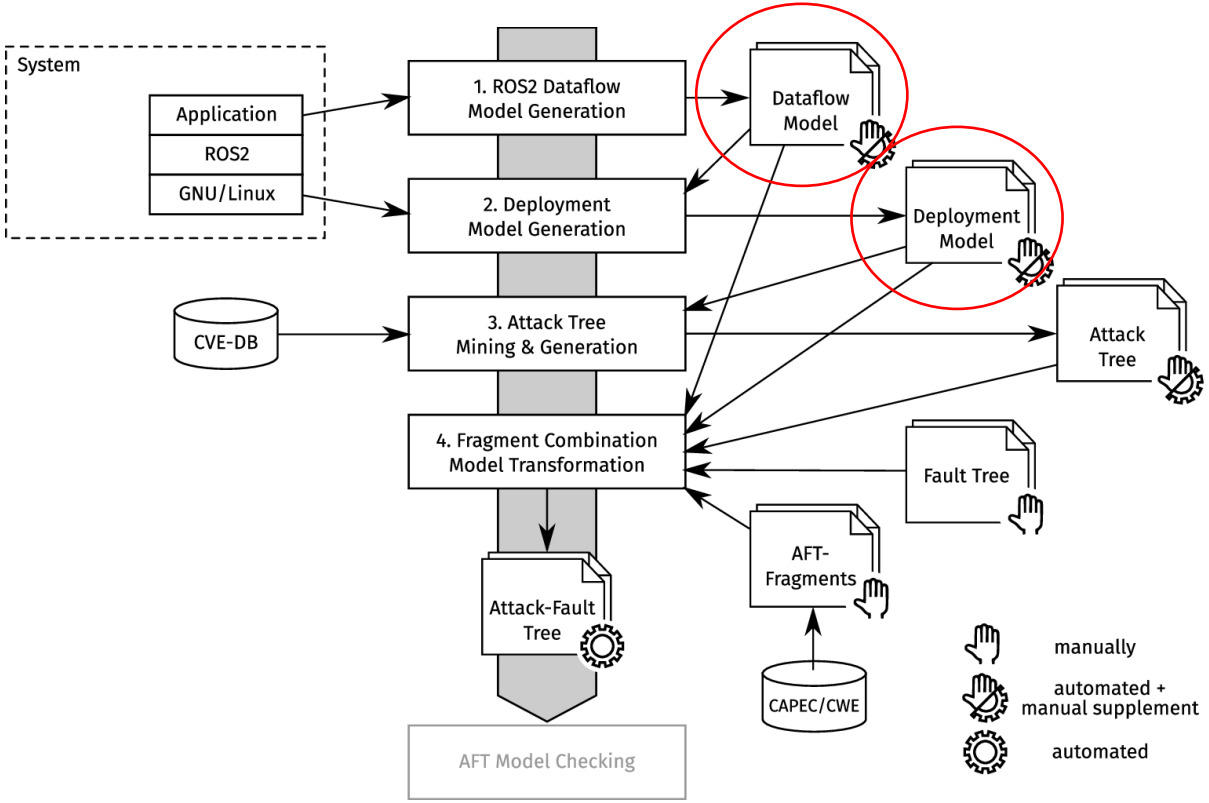
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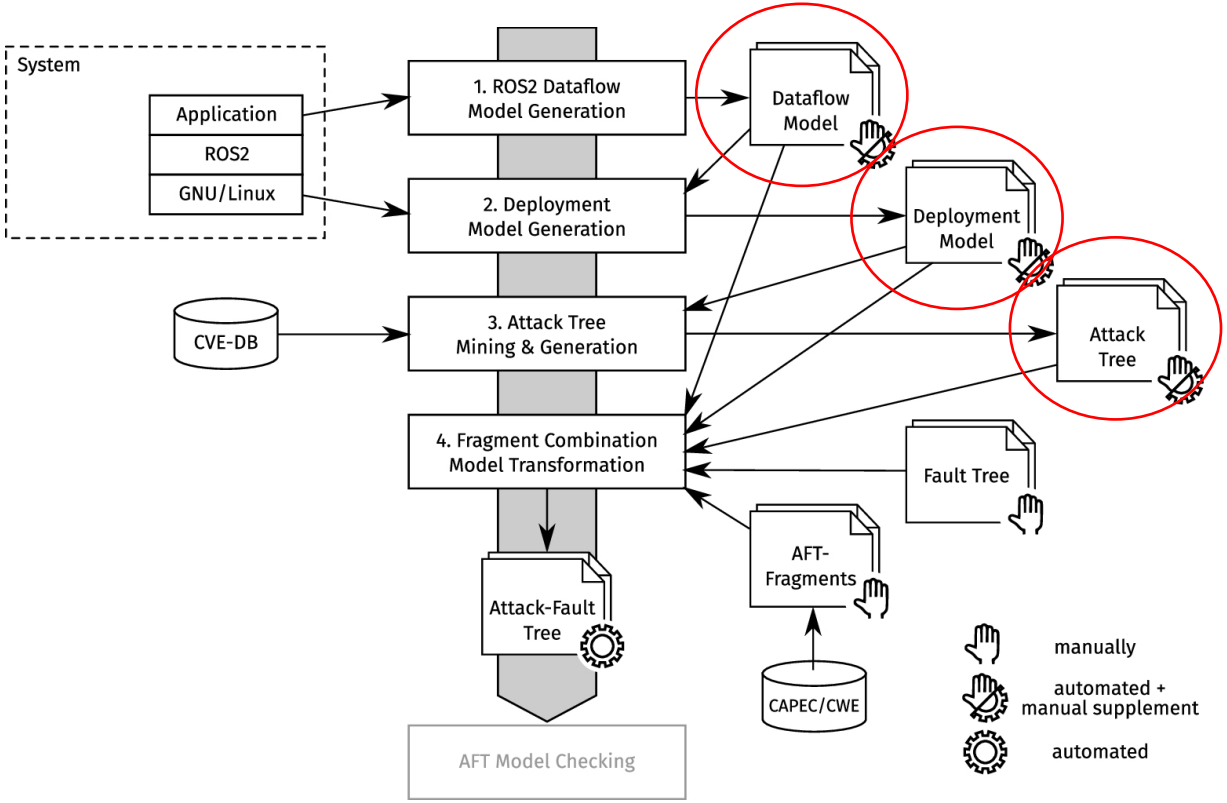
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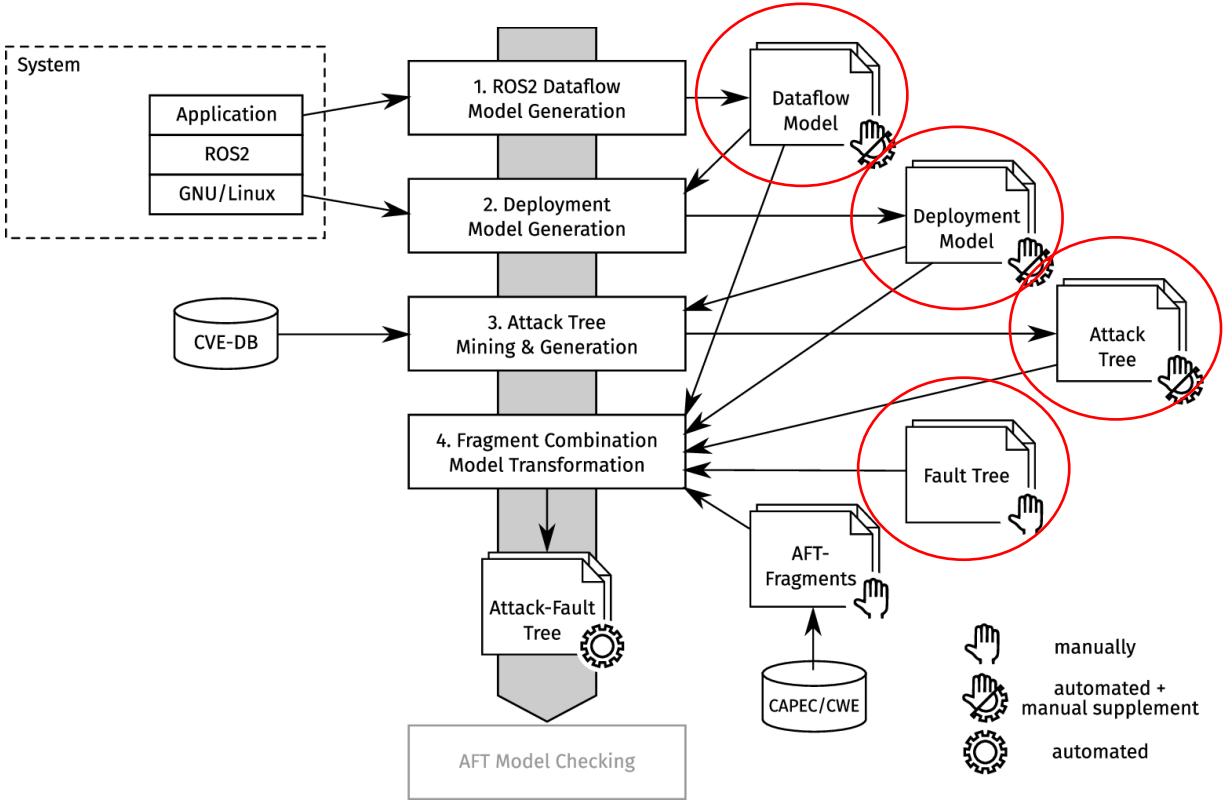
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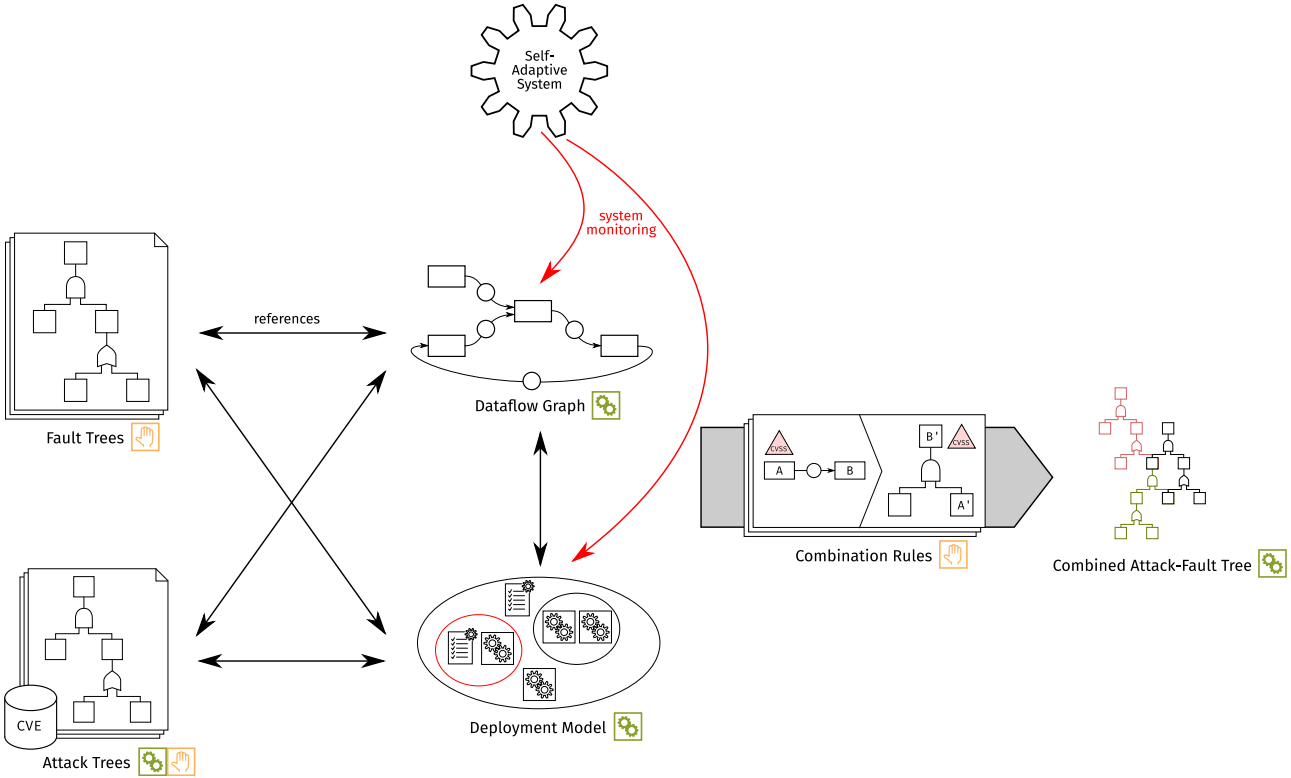
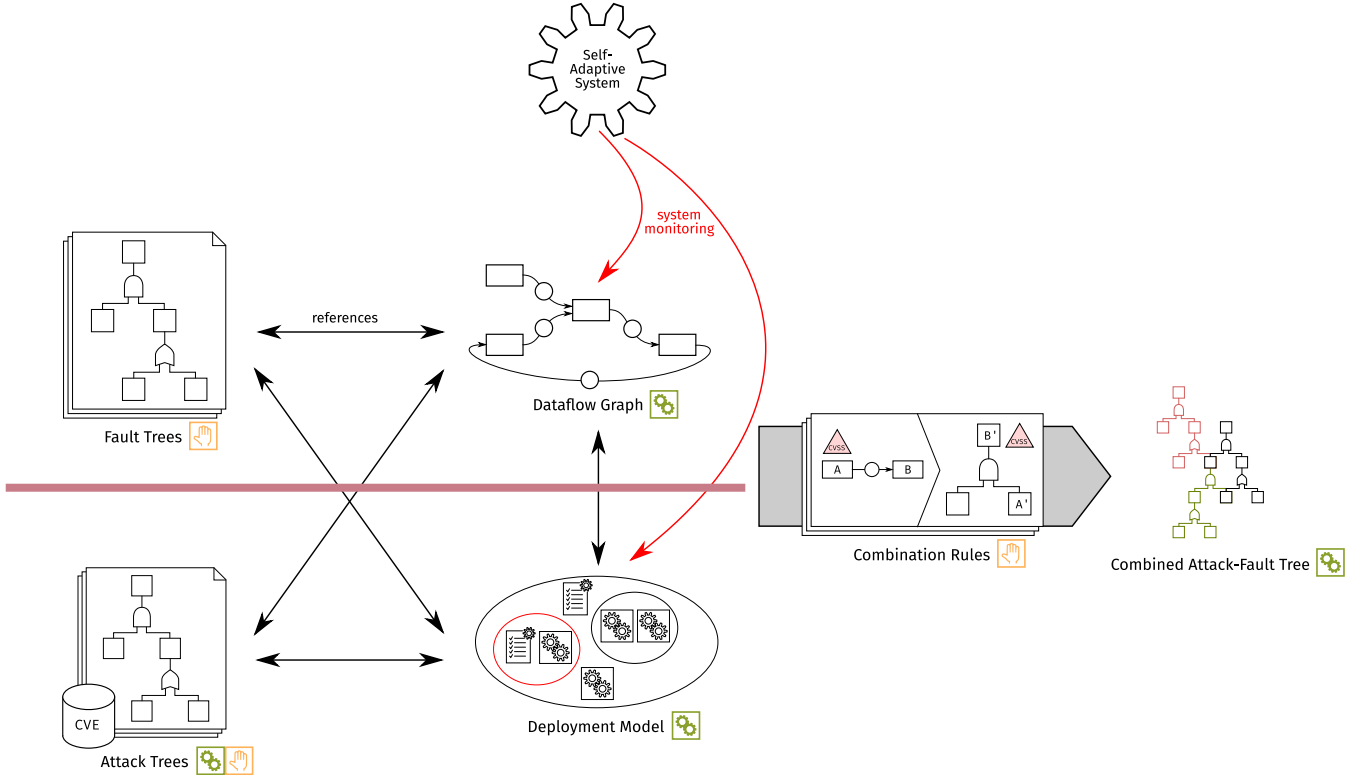
The SafeSec AFT Generation Toolchain (SAFT-GT)



Dataflow Model

The SafeSec AFT Generation Toolchain (SAFT-GT)



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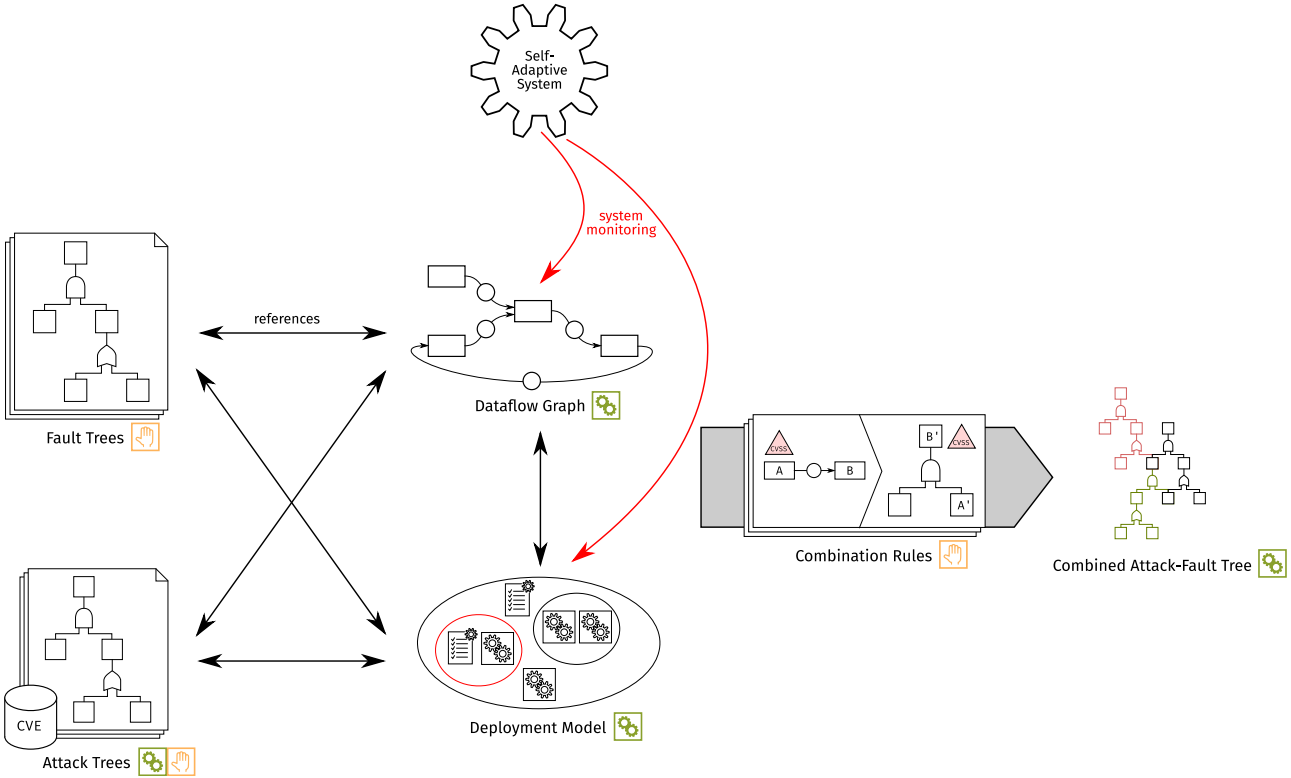

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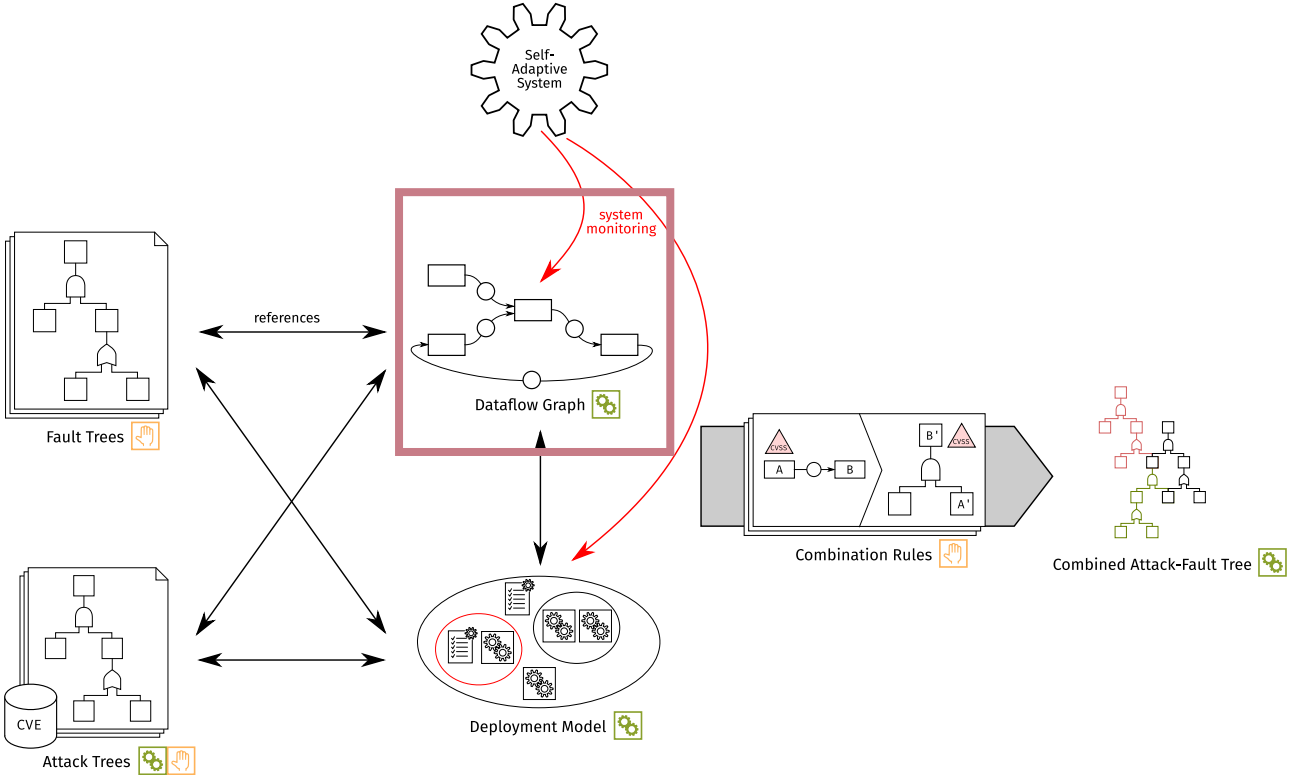
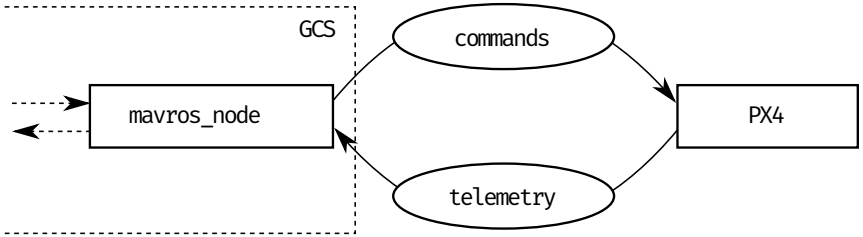


Dataflow Model

Dataflow Model

- Simple, only components and channels
- Logical view, independent from underlying realization/implementation



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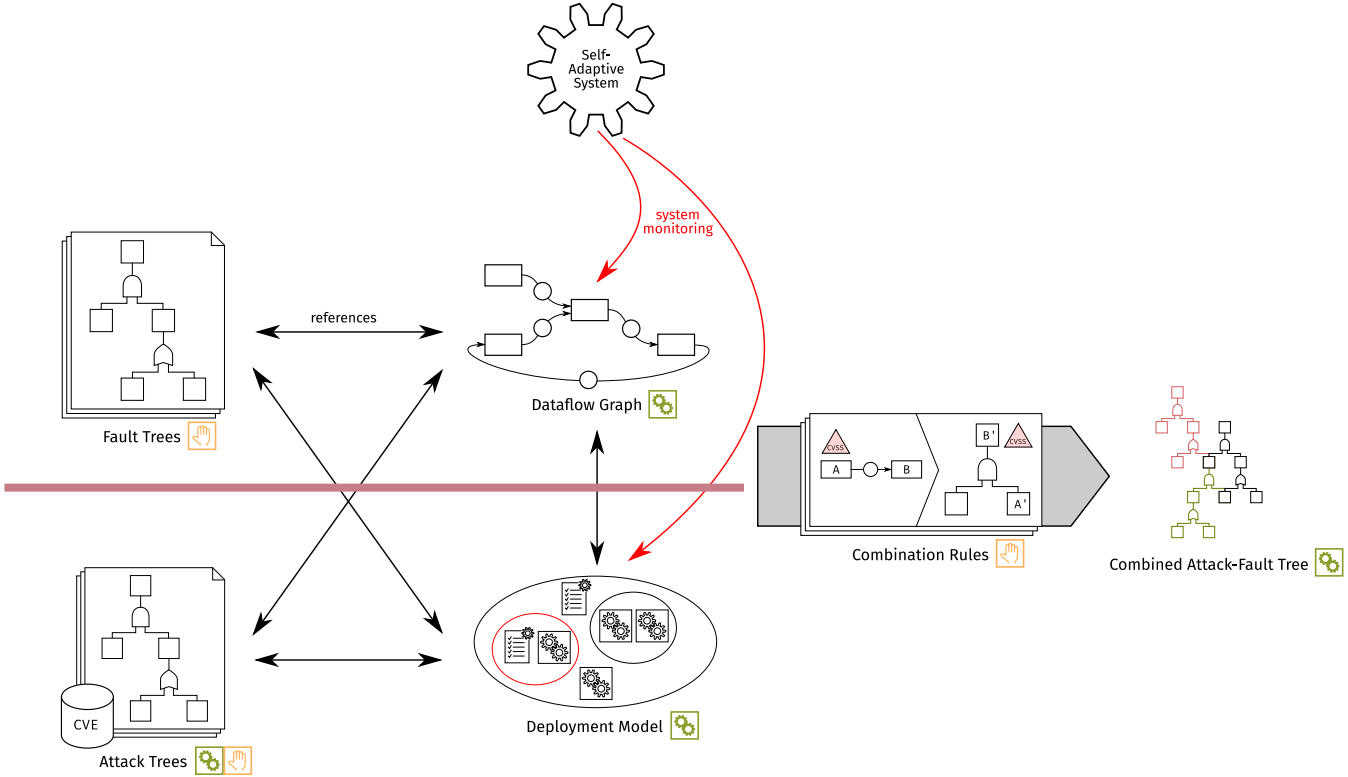

Dataflow Model → Deployment Model

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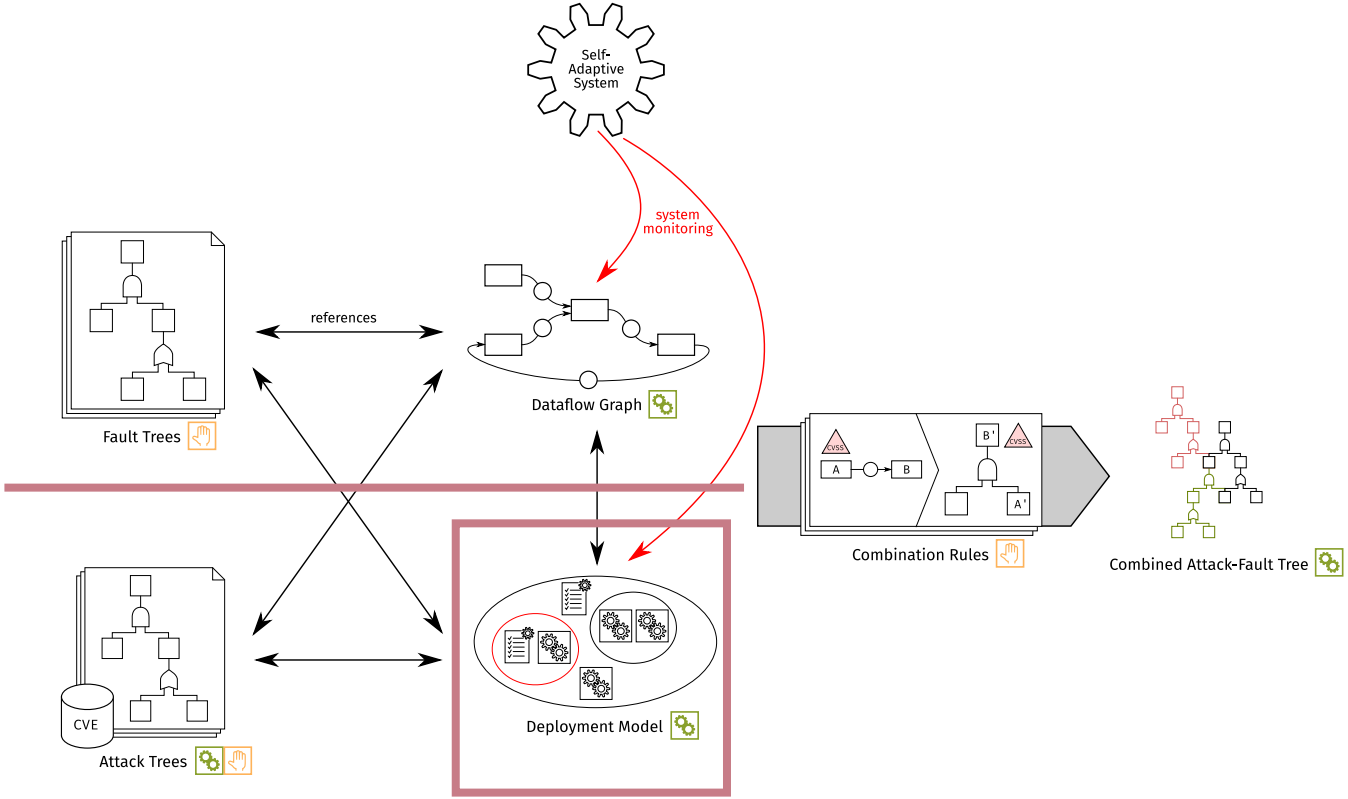

Dataflow Model → Deployment Model

Deployment Model

- Realization of components and channels
- Simple set-based representation, can be refined as needed

```
PX4:Component = {Mavlink1, pixhawk, ...}
Mavlink1:Protocol = {MavlinkLib, UDP, UART, ...}
commands:Channel = {Mavlink1, WiFi}
WiFi:Platform = {IEEE 802.11n, WPA2, UDP, TCP/IP, ...}
Radio:Platform = {UART}
mavros_node = {. . .} . . .
```

Dataflow Model | Deployment Model

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Dataflow Model → Deployment Model

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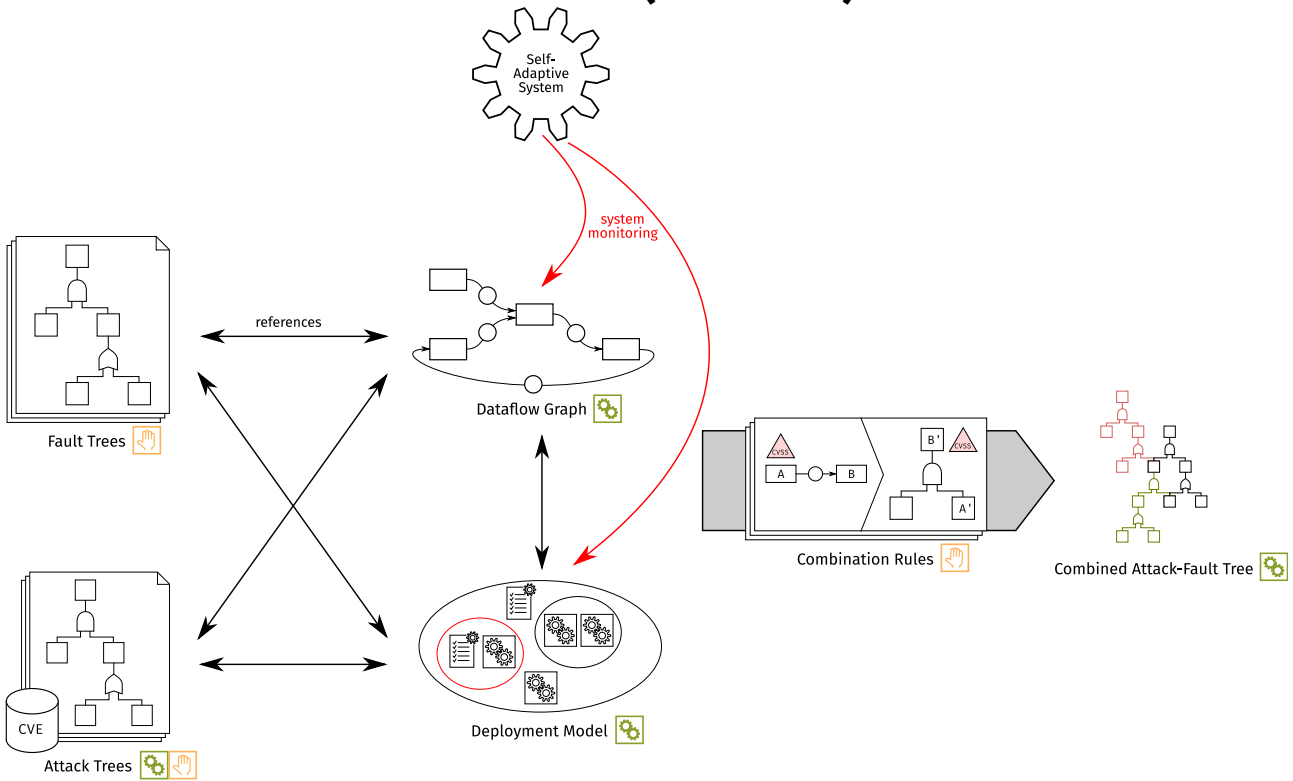

Dataflow Model → Deployment Model → Attack Tree

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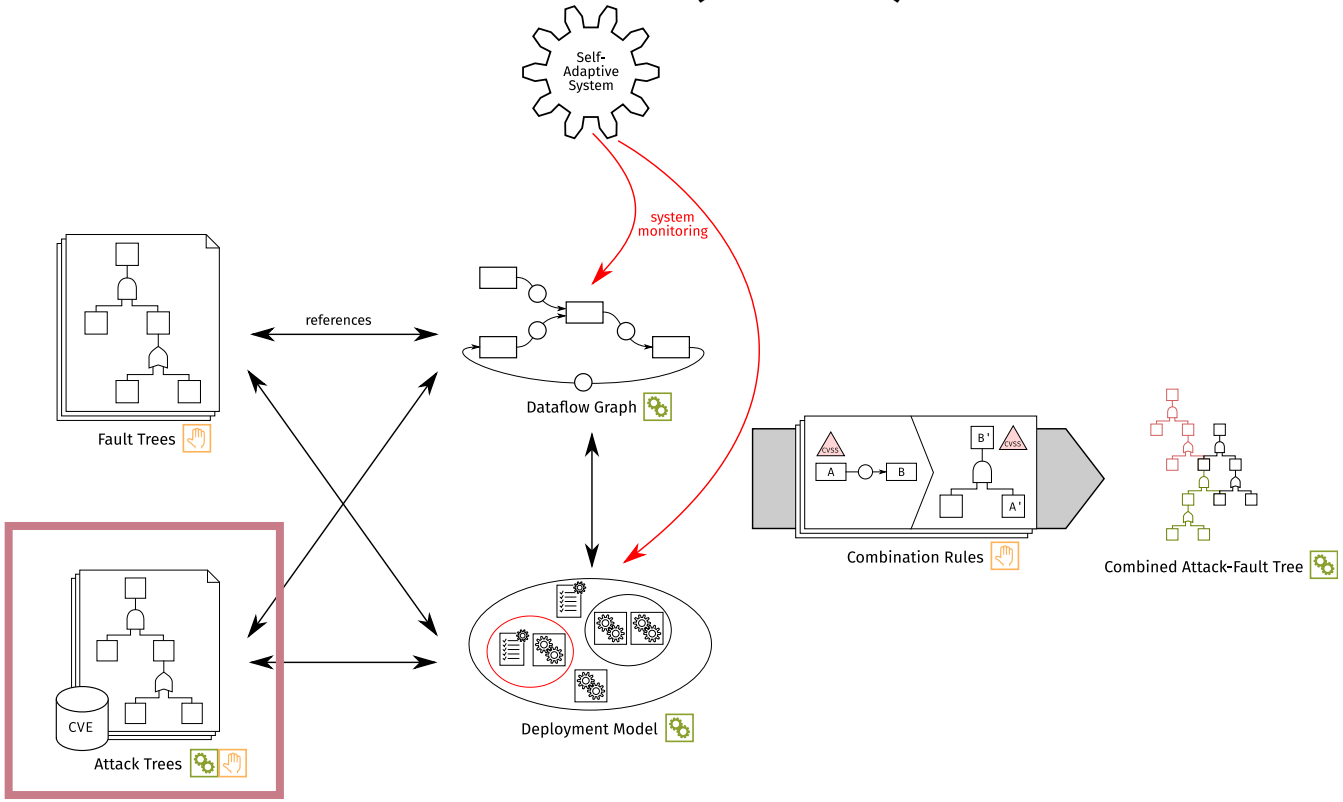
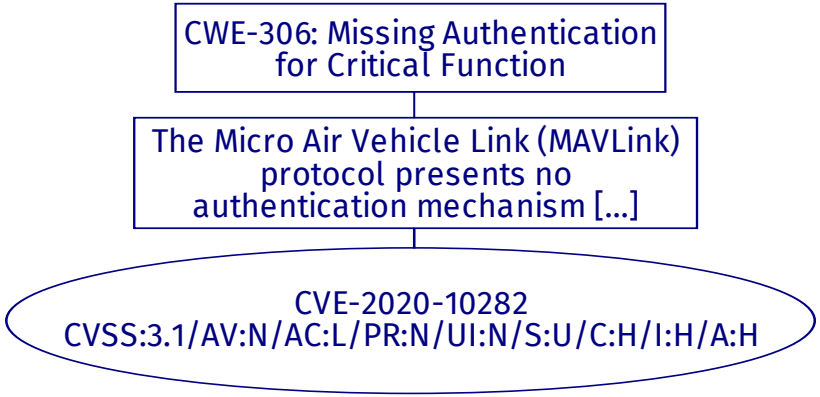


Dataflow Model → Deployment Model → Attack Tree

Attack Trees

Attack Trees

- Mined according to specific libraries/software (CPE)
- Use CVE, CVSS and EPSS data
- Often on the technical/platform level



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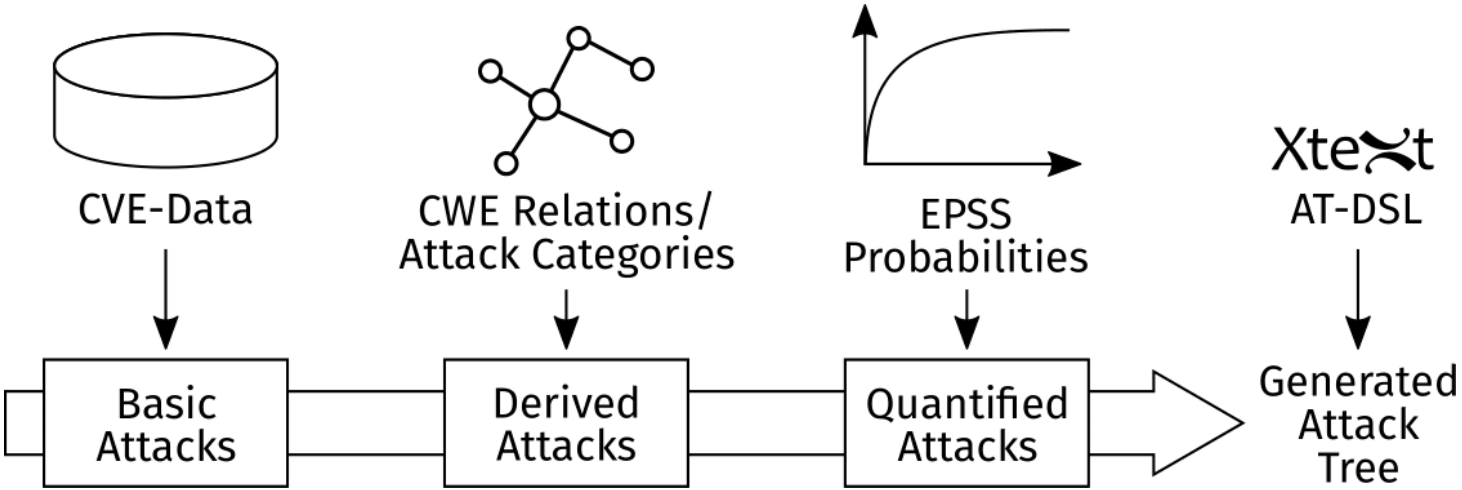

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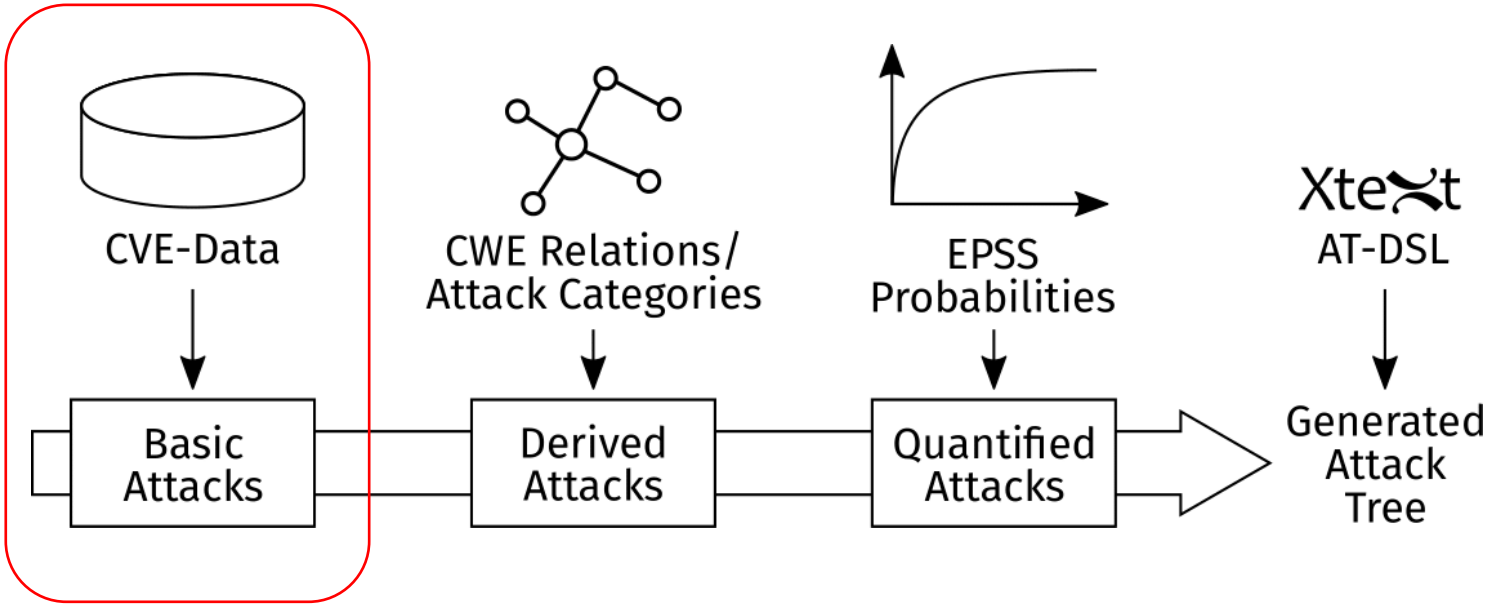

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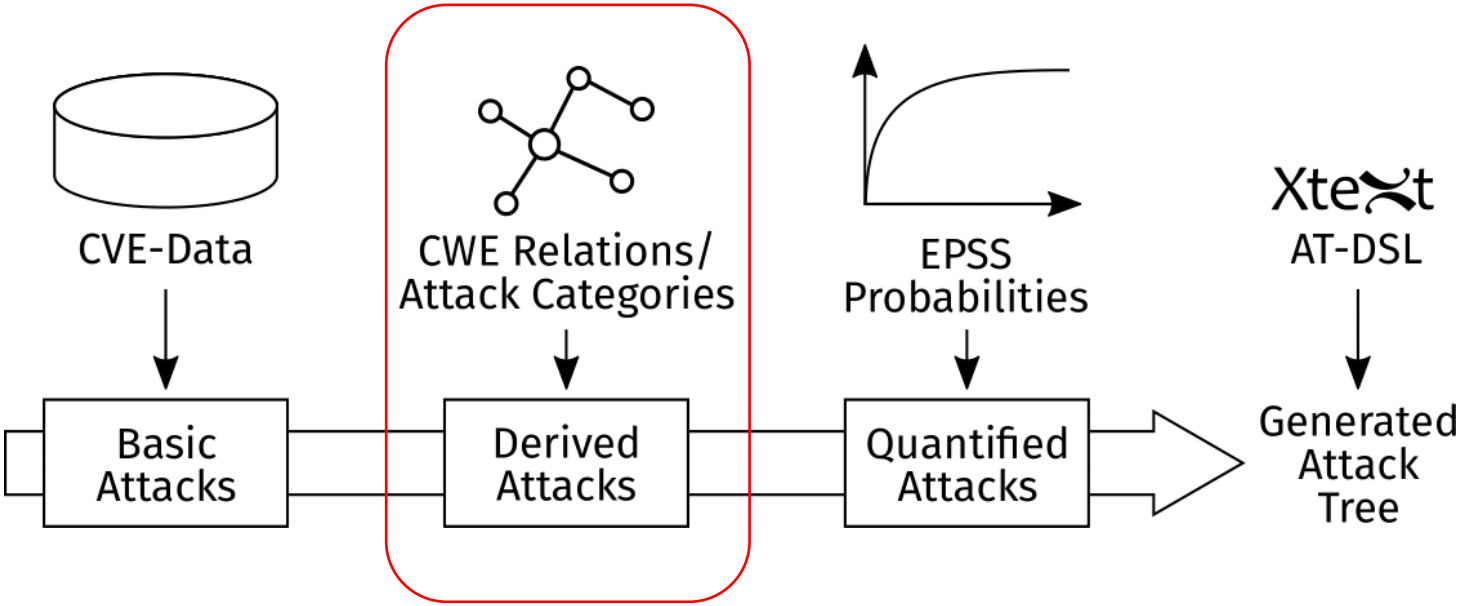

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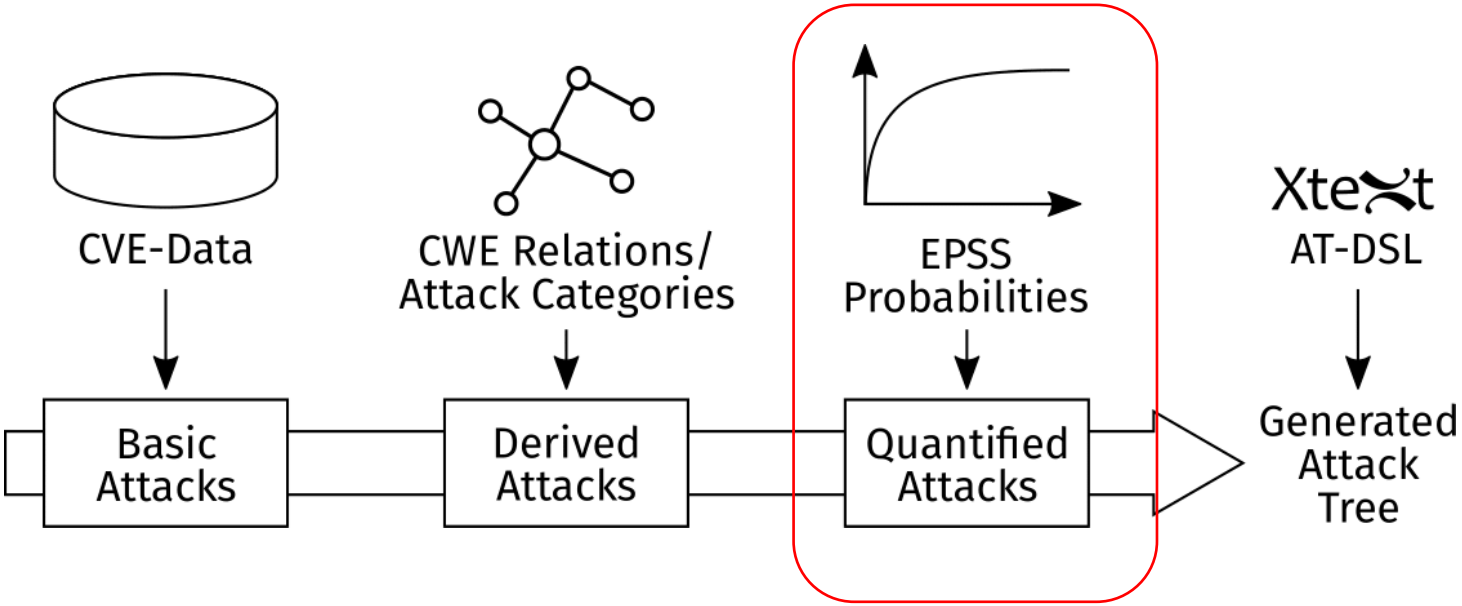

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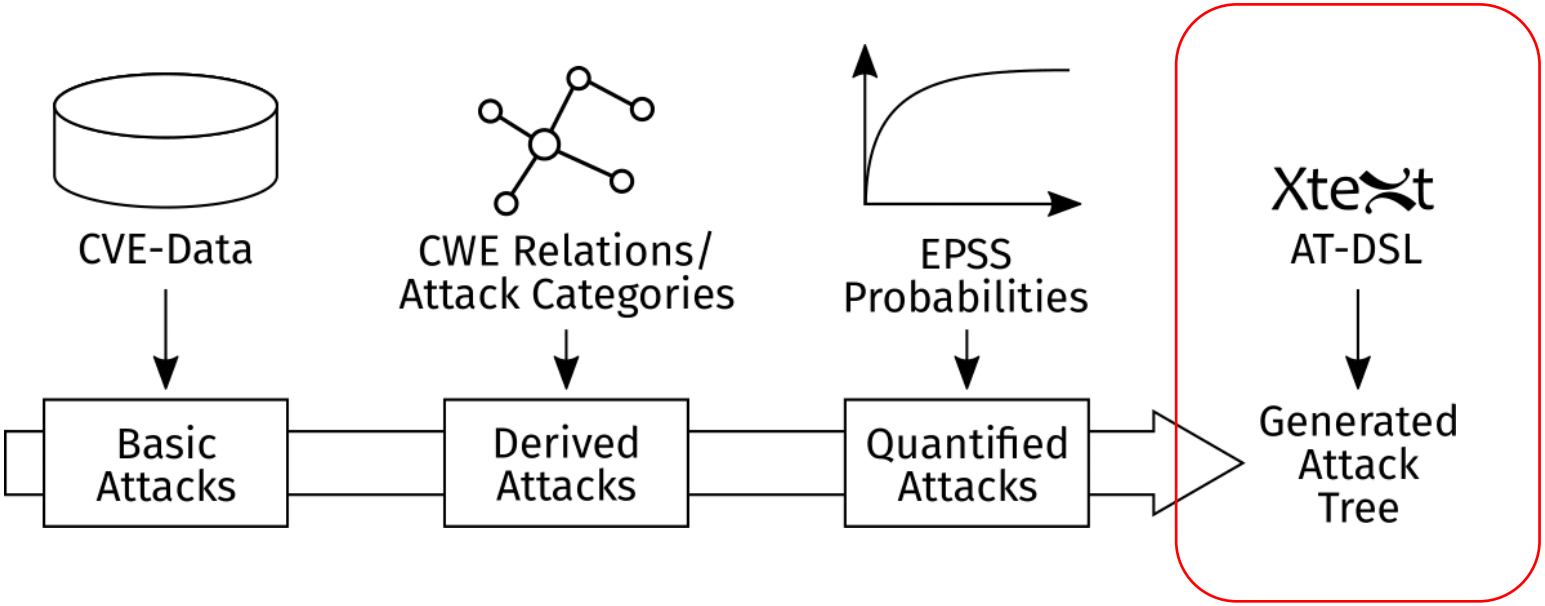

Basic Attack Tree

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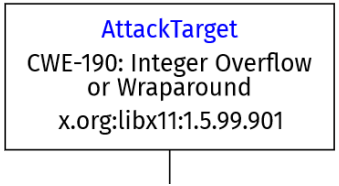

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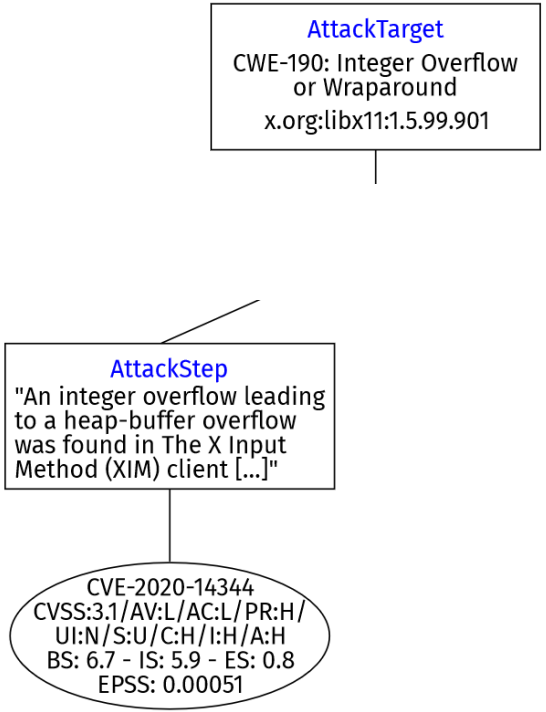

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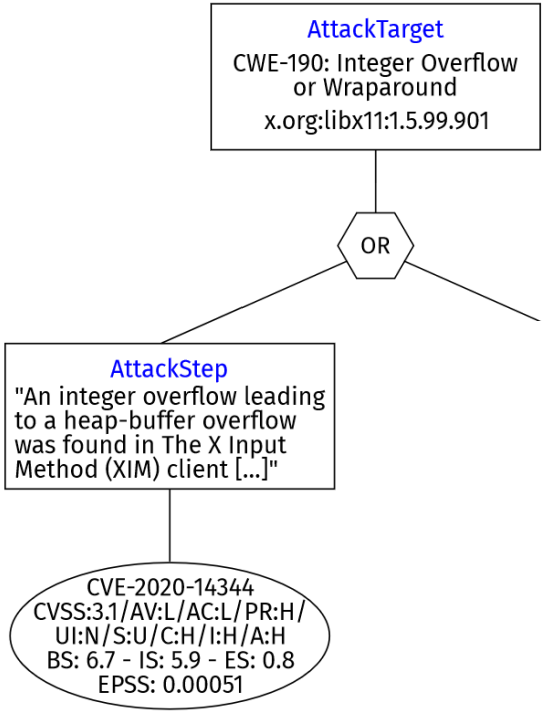

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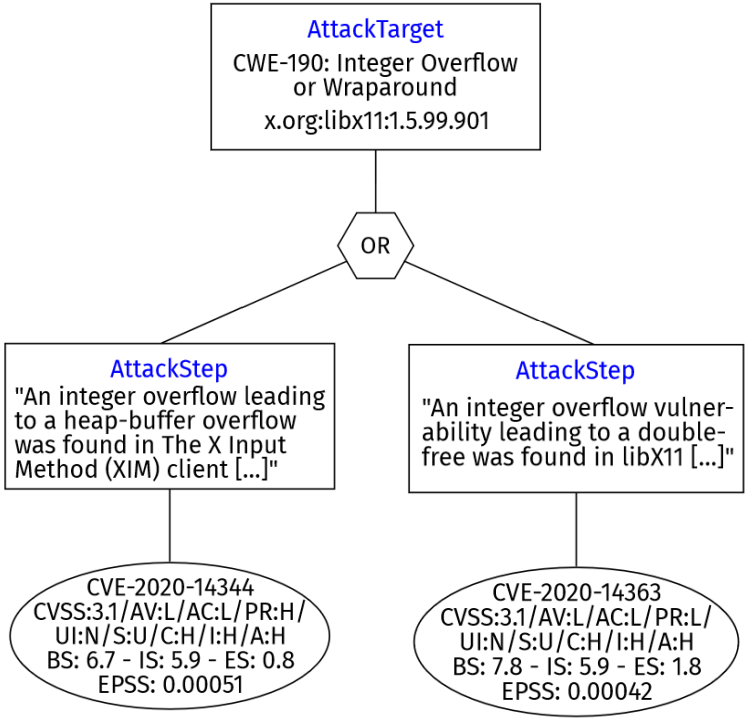

Derived Attack Tree

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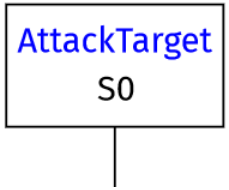

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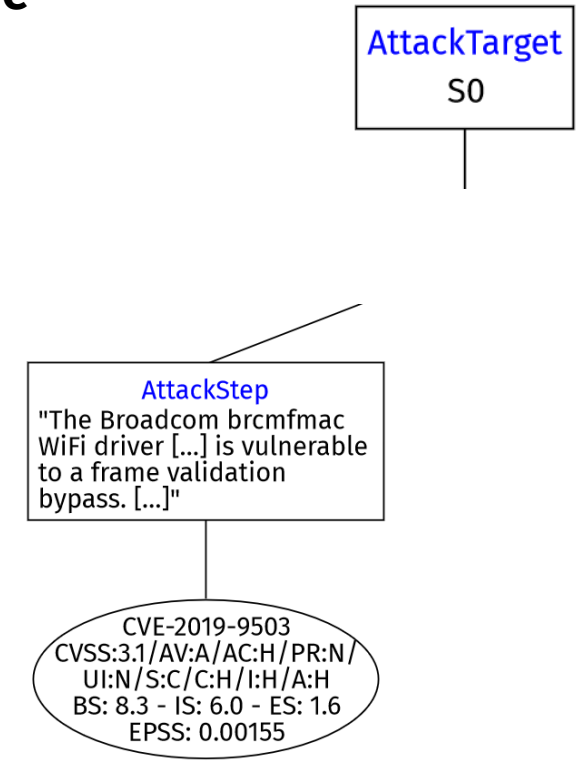

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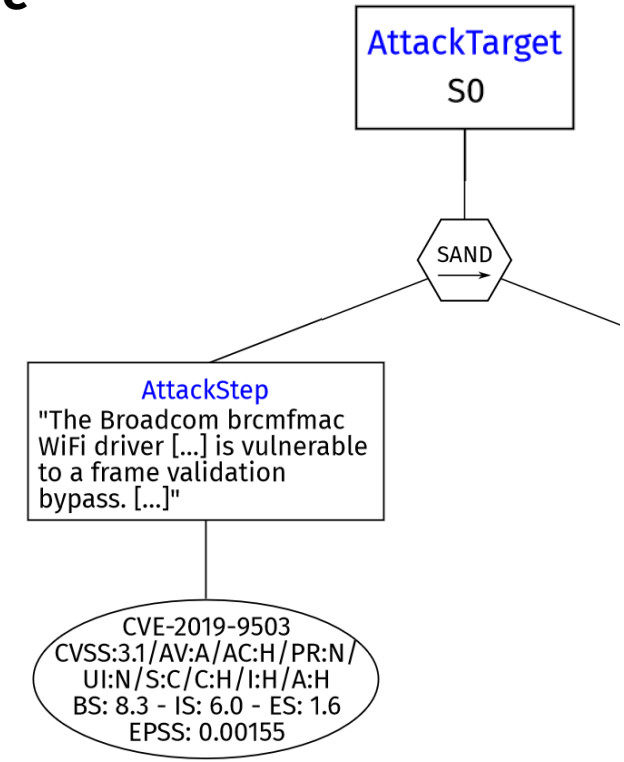

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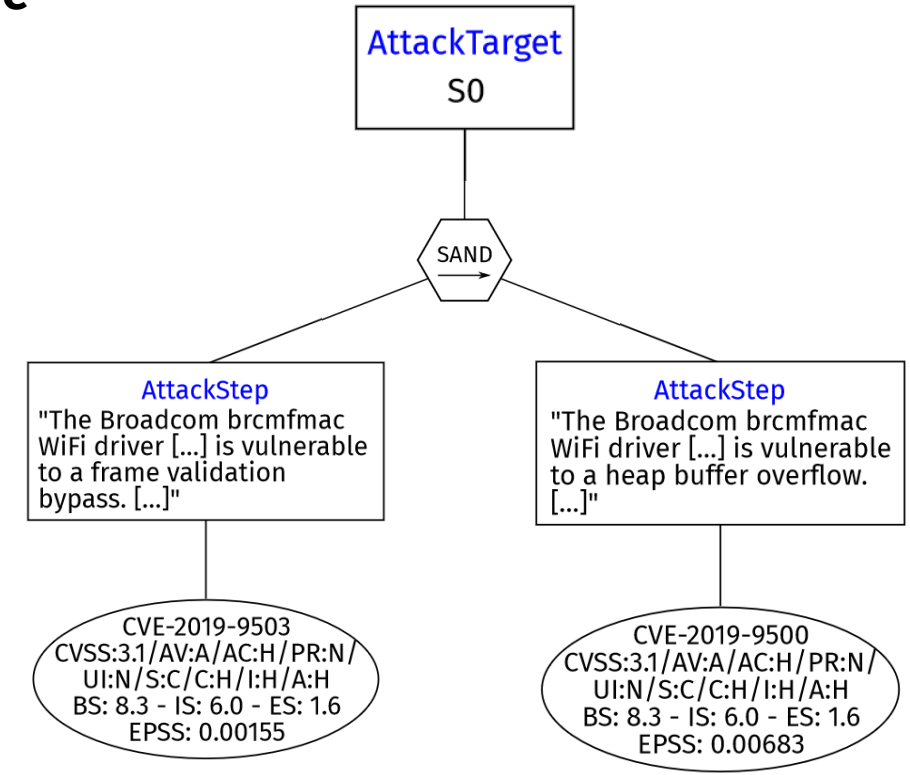

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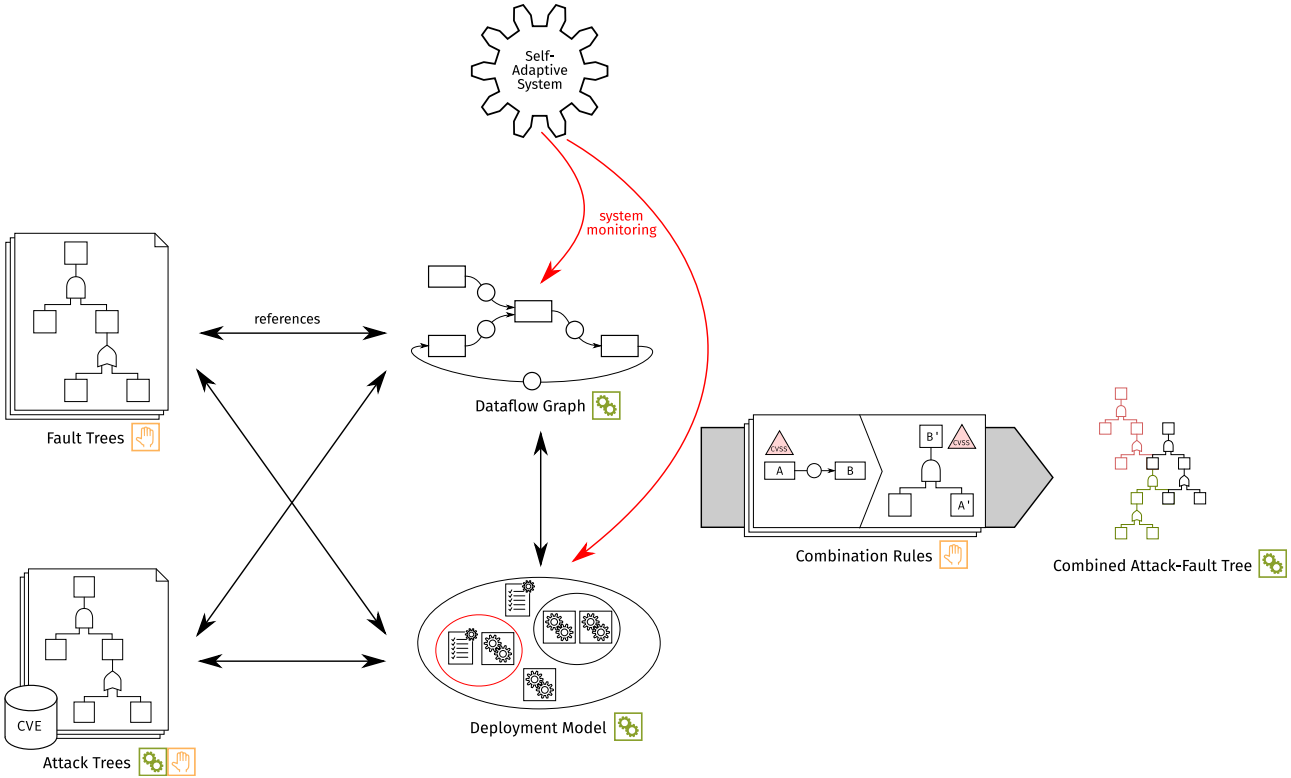

Dataflow Model → Deployment Model → Attack Tree → Fault Tree

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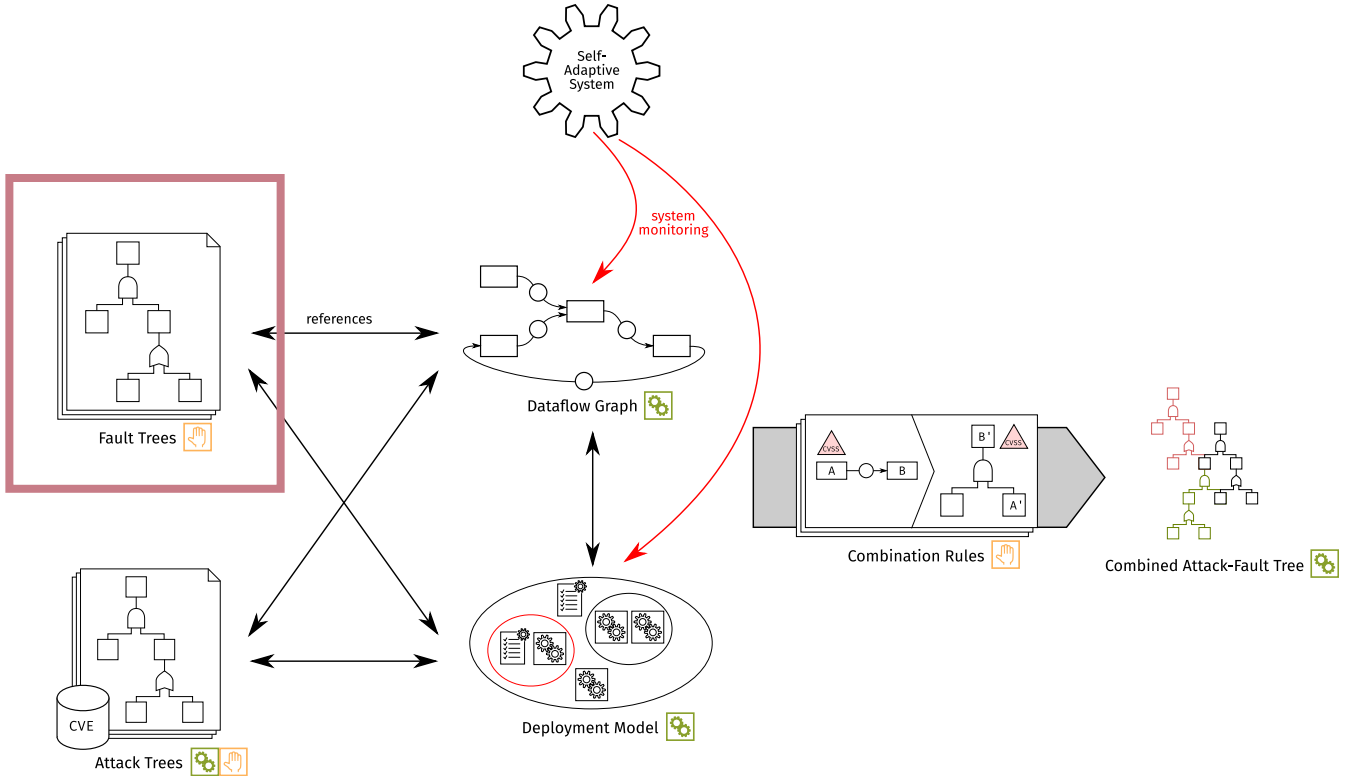

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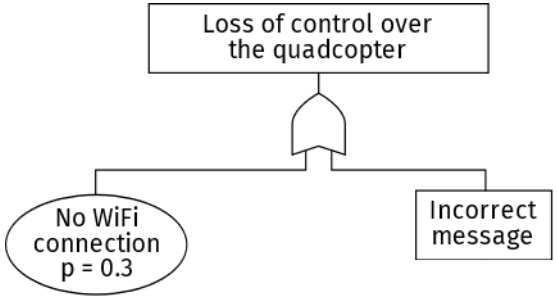
Fault Trees

Fault Trees

- Created by Domain experts (manually)
- On the logical/dataflow level



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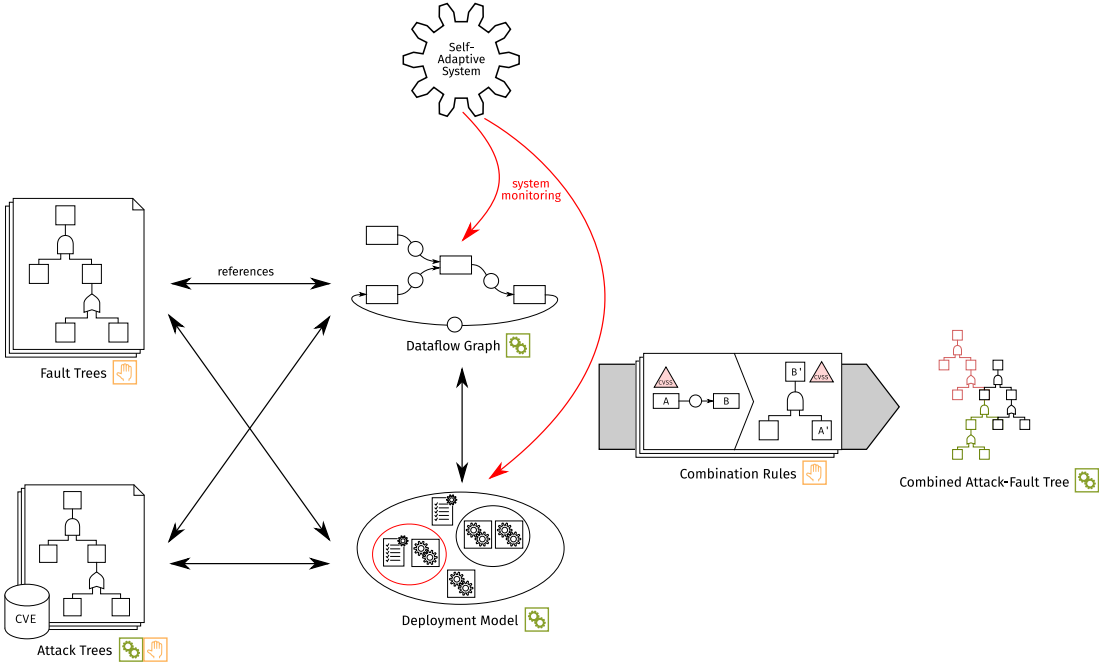
Any Questions?



→ Next up: Discussion on Safety/Security Analysis